

Metro Ticket Generating System in ServiceNow

1. Introduction

Project Title - Metro Ticket Generating System in ServiceNow

Team Member -Hema S

2. Project Overview

Purpose

The Metro Ticket Generating System is a ServiceNow-based application designed to digitize the metro ticket booking process. The system enables passengers to book tickets online, calculate fares automatically, generate QR-code-based tickets, and receive instant confirmations.

The project aims to:

- Reduce ticket counter queues.
- Improve commuter convenience.
- Promote cashless transactions.
- Automate fare calculation.
- Support paperless ticketing.

Features

- Metro Ticket Booking through Service Catalog.
- Source and Destination Selection.
- Passenger Type Selection.
- Automated Fare Calculation.
- QR Code Generation.
- Digital Ticket Creation.
- Email Notifications.
- Ticket History Tracking.
- Dashboard & Reports.
- Admin Management Portal.

3. Architecture

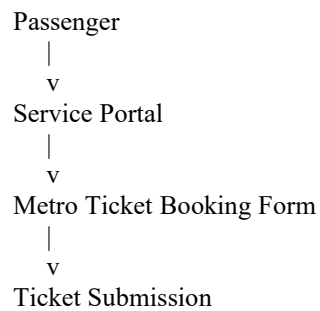
Frontend

The frontend is built using ServiceNow's user interface technologies:

Components

- Service Portal
- Service Catalog
- UI Policies
- Client Scripts
- UI Builder

User Flow



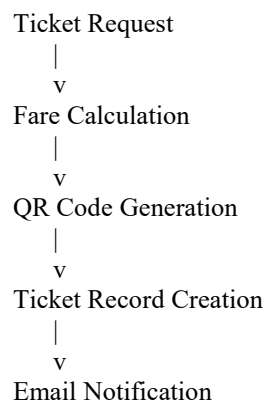
Backend

The backend logic is implemented using ServiceNow platform services.

Components

- Business Rules
- Flow Designer
- Script Includes
- Notifications
- Scheduled Jobs

Backend Flow



Database

Table: Metro Stations

u_metro_station

Field

Station Name

Station Code

Route

Status

Table: Fare Matrix

u_fare_matrix

Field

Source

Destination

Fare

Table: Metro Tickets

u_metro_ticket

Field

Ticket ID

Source

Destination

Passenger Type

Number of Tickets

Total Fare

QR Code

Journey Date

Status

4. Setup Instructions

Prerequisites

- ServiceNow Developer Instance
- Admin Access
- Service Portal Plugin
- Flow Designer Plugin
- Internet Connection

Installation

Step 1

Log in to ServiceNow Developer Instance.

Step 2

Create custom tables:

u_metro_station
u_fare_matrix
u_metro_ticket

Step 3

Create Service Catalog Item:

Metro Ticket Booking

Step 4

Configure:

- Client Scripts
- UI Policies
- Business Rules
- Flow Designer

Step 5

Configure QR Code API Integration.

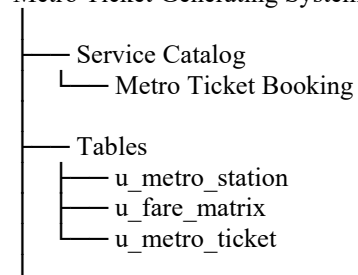
Step 6

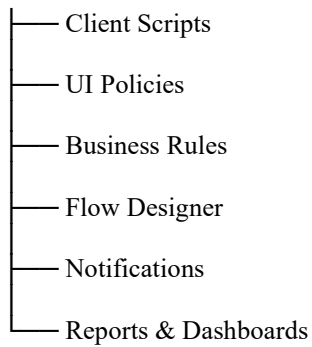
Test the application.

5. Folder Structure

Since ServiceNow stores components within the platform, the structure is:

Metro Ticket Generating System





6. Running the Application

Access Application

Service Portal

Steps

1. Open Service Portal.
2. Navigate to:

Metro Ticket Booking

1. Enter journey details.
2. Submit request.
3. Receive QR ticket.

7. API Documentation

Fare Calculation API

Method

POST

Request

```
{
  "source": "Ameerpet",
  "destination": "LB Nagar",
  "passengerType": "Adult",
  "tickets": "2"
}
```

Response

```
{  
  "fare": "80"  
}
```

QR Code Generation API

Method

POST

Request

```
{  
  "ticketId": "MT1001"  
}
```

Response

```
{  
  "qrCode": "Generated Successfully"  
}
```

8. Authentication

Authentication Method

ServiceNow User Authentication

Roles

Passenger

x_metro.passenger

Permissions:

- Book Ticket
- View Ticket History

Admin

x_metro.admin

Permissions:

- Manage Stations
- Manage Fare Matrix
- View Reports

Security Features

- Role Based Access Control (RBAC)
- ACL Rules
- HTTPS Communication
- Secure Record Access

9. User Interface

Screens

Home > Service Catalog > Office > Services > Book A Metro Ticket

Search Catalog

Book A Metro Ticket

Book A Metro Ticket

* Indicates required

* Enter Smart card number

Where do you want to go?

☒ Chennai

☐ Coimbatore

☐ Salem

Type of Journey

☒ None

☐ Return[add \$200.00]

☐ Single[add \$200.00]

☐ Up and Down[add \$400.00]

Starting from

Quantity: 1

Price: \$100.00

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

Required information

Enter Smart card number

Metro Ticket Booking Form

Fields:

- Smart Card Number
- Destination
- Type of Journey
- Number of Tickets

Screen 2

Home > Request Summary - REQ0010003

Search Catalog

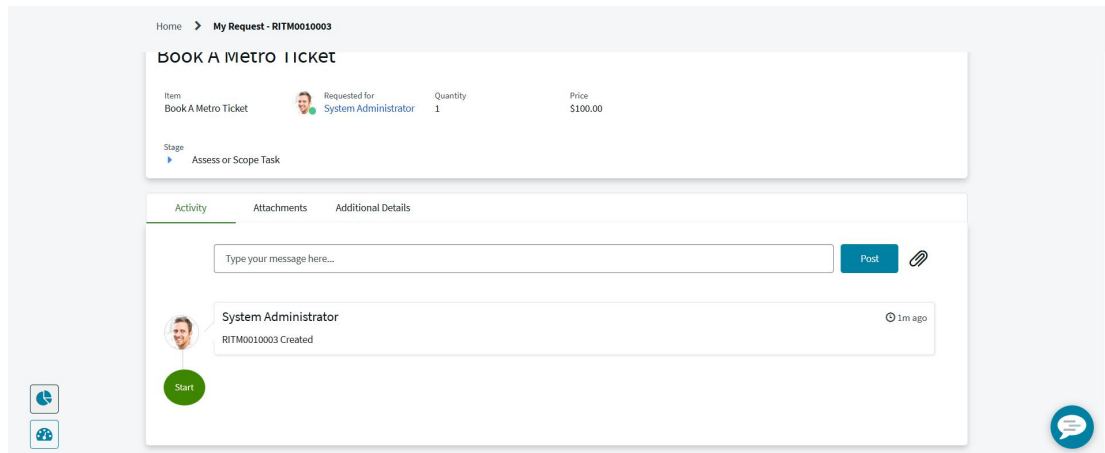
Submitted :2026-06-28 02:06:45
Request Number : REQ0010003
Estimated Delivery : 2026-06-30

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Book A Metro Ticket	2026-06-30	Assess or Scope Task	\$100.00	1	\$100.00
					Total: \$100.00

Displays:

- Fare
- Total Amount

Screen 3



Displays:

- Booking Details

10. Testing

Testing Strategy

Unit Testing

- Client Scripts
- Business Rules

Integration Testing

- QR Code API
- Notifications

System Testing

- End-to-End Ticket Booking

User Acceptance Testing

- Passenger Validation
- Admin Validation

Test Result

Total Test Cases : 60
Passed : 59
Failed : 1
Pass Percentage : 98.33%

11. Screenshots

The first screenshot shows the 'Metro Ticketing System' app details in ServiceNow Studio. The app is a custom application with scope 'x_1971522_metro_0', created on 2026-06-22 15:27:15. The left sidebar shows the 'Filter list' with categories like Security, Role (2), and Other. The main area displays a table of files with columns: Name, Type, Updated, Updated by, and Created by. The table is currently empty, with instructions for adding data, user interface, automation, or security.

The second screenshot shows the 'Now, let's get more info about your new table' screen. It prompts the user to define the properties of the new table. The 'Table label' is 'Metro Station's Details'. The 'Table name prefix' is 'x_1971522_metro_0'. The 'Table name' is 'metro_station_s_details'. There are checkboxes for 'Make extensible' and 'Auto number', both of which are currently unchecked. The 'Continue' button is visible at the bottom right.

The third screenshot shows the 'CREATE TABLE' screen. It displays a progress bar with three steps: Source, Define, and Summary. The 'Source' step is selected. The table is named 'Metro Station's Details' and has 0 fields. The 'Edit table' and 'Done' buttons are visible at the bottom right.

[illegible]

servicenow

AllFavoritesHistoryWorkspaces

Catalog Item - Book A Metro Ticket Item

Ap...Up...Search

CopyTry ItUpdateEdit in Catalog BuilderDelete

Catalog ItemBook A Metro Ticket Item

Press Alt+0 for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.

Meta

CopyTry ItUpdateEdit in Catalog BuilderDelete

Related Links

[Item Diagnostic](#)
[Run Point Scan](#)
[\[SN Utils\] Versions \(4\)](#)

VariablesVariable SetsCatalog UI Policies (1)Catalog Client Scripts (1)Available ForNot Available ForCategories (1)Catalogs (1)Catalog Data Lookup DefinitionsRelated ArticlesRelated Catalog Items

Assigned Topics

OrderSearch

Actions on selected rows...New

Catalog item = Book A Metro Ticket Item

	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	Book A Metro Ticket Item	(empty)		true	true	false	2026-06-27 11:11:13	100

1 to 1 of 1

CopyTry ItUpdateEdit in Catalog BuilderDelete

Related Links

[Item Diagnostic](#)
[Show VA render type](#)
[Run Point Scan](#)
[\[SN Utils\] Versions \(8\)](#)

VariablesVariable SetsCatalog UI Policies (1)Catalog Client Scripts (3)Available ForNot Available ForCategories (1)Catalogs (1)Catalog Data Lookup DefinitionsRelated ArticlesRelated Catalog Items

Assigned Topics

for textSearch

Actions on selected rows...New

Catalog item = Book A Metro Ticket Item

	Name	Active	Table	View	Type	Updated
☐	Validation	true			onChange	2026-06-28 00:36:15
☐	QR Generation	true			onSubmit	2026-06-28 00:34:13
☐	FareCalculation	true			onChange	2026-06-27 10:58:15

1 to 3 of 3

Catalog ItemBook A Metro Ticket Item

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

NameBook A Metro Ticket Item

ApplicationMetro Ticketing System

CatalogsService Catalog

CategoryServices

State-- None --

Checked out-- None --

OwnerSystem Administrator

Active

Fulfillment automation levelUnspecified

Item DetailsProcess EnginePicturePricingPortal Settings

Short description

Description

Catalog ItemNew record

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Item DetailsProcess EnginePicturePricingPortal Settings

Short descriptionBook A Metro Ticket

Description

Conclusion

The **Metro Ticket Generating System in ServiceNow** provides a secure, automated, and paperless ticketing solution that improves passenger convenience, increases operational efficiency, promotes digital payments, and supports data-driven decision-making for metro transportation services.